

The half-standard-error pooling rule for interactions is stated in the wrong standard error

An exact computation answering catalog problem MOD-15

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Abstract

Background. An individual-data network meta-analysis has within-trial and across-trial information about a treatment-covariate interaction. The only published criterion for pooling them is that the within-trial estimate lie within half a standard error of the across-trial estimate (1), described as relatively strict and never calibrated.

Methods. Under equality the difference of the two estimates has standard error $\sqrt{s_W^2 + s_A^2}$, not s_A , so the rule’s operating characteristics are bivariate-normal integrals. We computed them exactly for 24 network configurations (5, 10 or 20 trials of 200; trial covariate means spread with SD 0.25 or 0.75; ecological bias 0 to 0.4), and checked the standard errors and two exact quantities against 2000 individual-level networks.

Results. The rule pools with probability at most $P(|Z| < 0.5) = 0.383$ under equality; in the configurations studied 0.314 to 0.374. When it does pool under ecological bias, the pooled interval’s coverage falls to 0.221. Its RMSE is within 0.3% of never pooling in every configuration, so it forgoes the 2% to 19% RMSE reduction pooling offers when the two estimates agree.

Conclusion. The rule is strict in every regime, not only some, and it is strict in the wrong way: it refuses pooling most of the time when pooling is valid, and when it admits pooling under bias the interval is not trustworthy. A test on the difference in its own standard error is the minimal repair; no rule of this kind can detect bias small relative to the across-trial standard error.

1 The problem

Across-trial interaction information is identified only if there is no trial-level confounding of the association between trial covariate means and trial effects (2,3). Pooling it with the within-trial estimate gains precision at the risk of ecological bias δ . Write $\hat{\beta}_W \sim N(\beta, s_W^2)$ and $\hat{\beta}_A \sim N(\beta + \delta, s_A^2)$, independent. The rule pools when $|D| < \frac{1}{2}s_A$ with $D = \hat{\beta}_W - \hat{\beta}_A \sim N(-\delta, s_W^2 + s_A^2)$. Under $\delta = 0$,

$$P(\text{pool}) = P\left(|Z| < \frac{0.5 s_A}{\sqrt{s_W^2 + s_A^2}}\right) \leq 0.383,$$

approached when $s_A \gg s_W$. The catalog’s design argued the rule is “not necessarily strict” and loosest with few trials. It is strict everywhere; the ratio s_W/s_A only decides how strict.

2 Methods

Registered protocol: `protocol.md`. Standard errors came from a network of K trials of 200 (1:1), within-trial covariate SD 1, residual SD 1: $s_W^2 = 4/(200K)$ and $s_A^2 = (4/200)/\sum_k (\bar{x}_k - \bar{x})^2$. Conditional on D , the within-trial error is normal with mean $s_W^2(D + \delta)/(s_W^2 + s_A^2)$, so the error of whatever a policy reports, and its interval's coverage, integrate exactly over D . Policies: never pool; always pool; the half- s_A rule; tests on D at size 5% and 20%. The individual-level check (`R/01-check.R`, 2000 networks per row) reproduced the standard errors and the exact pass probability and conditional coverage, for example a pass rate of 0.366 against 0.374 exact.

3 Results

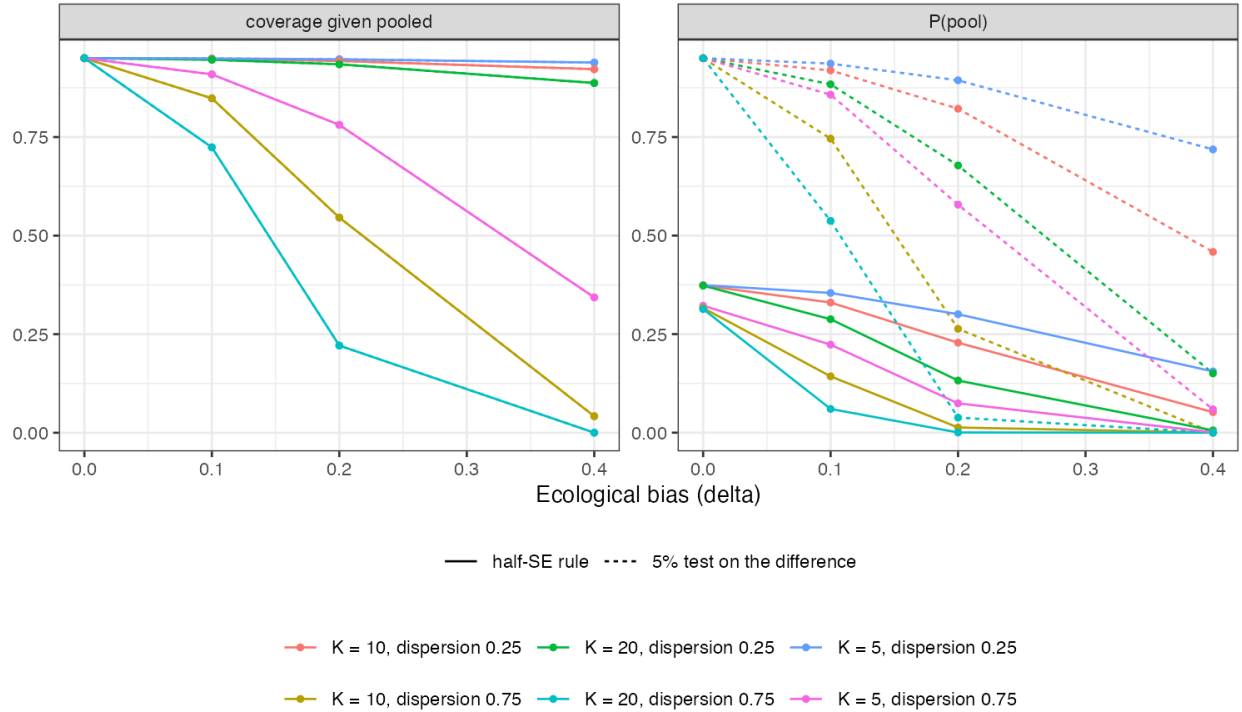


Figure 1: Probability of pooling and coverage of the pooled interval given pooling, by ecological bias.

Table 1: The half-SE rule by configuration and ecological bias.

K	dispersion	delta	sW/sA	P(pool)	coverage given pooled	RMSE
5	0.25	0.0	0.224	0.374	0.950	0.063
10	0.25	0.0	0.237	0.373	0.950	0.045
20	0.25	0.0	0.244	0.373	0.950	0.032
5	0.75	0.0	0.671	0.322	0.950	0.063
10	0.75	0.0	0.712	0.316	0.950	0.045
20	0.75	0.0	0.731	0.314	0.950	0.032
5	0.25	0.1	0.224	0.354	0.949	0.063
10	0.25	0.1	0.237	0.330	0.948	0.045
20	0.25	0.1	0.244	0.288	0.946	0.032
5	0.75	0.1	0.671	0.223	0.909	0.063

Table 1: The half-SE rule by configuration and ecological bias.

K	dispersion	delta	sW/sA	P(pool)	coverage given pooled	RMSE
10	0.75	0.1	0.712	0.143	0.848	0.045
20	0.75	0.1	0.731	0.060	0.724	0.032
5	0.25	0.2	0.224	0.301	0.947	0.063
10	0.25	0.2	0.237	0.228	0.943	0.045
20	0.25	0.2	0.244	0.132	0.935	0.032
5	0.75	0.2	0.671	0.074	0.781	0.063
10	0.75	0.2	0.712	0.013	0.546	0.045
20	0.75	0.2	0.731	0.000	0.221	0.032
5	0.25	0.4	0.224	0.155	0.939	0.063
10	0.25	0.4	0.237	0.052	0.922	0.045
20	0.25	0.4	0.244	0.006	0.887	0.032
5	0.75	0.4	0.671	0.001	0.343	0.063
10	0.75	0.4	0.712	0.000	0.042	0.045
20	0.75	0.4	0.731	0.000	0.000	0.032

The rule pools in about a third of analyses where pooling is valid (Table 1). As ecological bias grows, pooling becomes rarer but not rare enough: at 20 trials with dispersed trial means and $\delta = 0.2$ it pools in 0.000 of analyses, and the pooled interval then covers in 0.221 (Figure 1). Averaged over pooling and not pooling, the rule behaves like never pooling: its RMSE matches never pooling to within 0.3%. The size-5% test pools more often under equality and fails similarly under bias; always pooling has the lowest RMSE without bias and the worst with it. Never pooling had the lowest worst-case RMSE over $\delta \in [0, 0.2]$ in five of six network configurations.

4 What this does not answer

The two estimates are treated as independent and normal with known standard errors; between-trial heterogeneity in the treatment effect, which widens s_A , is not modeled, and the network is IPD in every trial. The ML-NMR case, where aggregate trials inform only the across-trial component, is not computed. Nothing here tests whether trial-level confounding is present; no agreement rule can. Peer review has not been done.

References

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